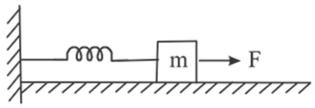
JEE Mains 2019 Chapter wise Question Bank

SHM - Questions

Q1

A block of mass m, lying on a smooth horizontal surface, is attached to a spring (of negligible mass) of spring constant k. The other end of the spring is fixed, as shown in the figure. The block is initally at rest in its equilibrium position. If now the block is pulled with a constant force F, the maximum speed of the block is:



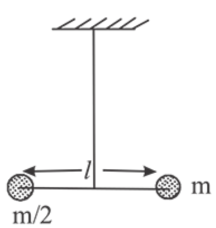
(1) $\frac{2F}{\sqrt{mk}}$ (2) $\frac{F}{\pi\sqrt{mk}}$

(3) $\frac{\pi F}{\sqrt{mk}}$ (4) $\frac{F}{\sqrt{mk}}$

9 Jan Morning

Q2

Two masses m and $\frac{m}{2}$ are connected at the two ends of a massless rigid rod of length l. The rod is suspended by a thin wire of torsional constant k at the centre of mass of the rod-mass system (see figure). Because of torsional constant k, the restoring toruque is $\tau = k\theta$ for angular displacement θ. If the rod is rotated by θ_0 and released, the tension in it when it passes through its mean position will be:



(1) $\frac{3k\theta_0^2}{l}$ (2) $\frac{2k\theta_0^2}{l}$

(3) $\frac{k\theta_0^2}{l}$ (4) $\frac{k\theta_0^2}{2l}$

9 Jan Morning

Q3

A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to :

(1) 0.77 (2) 0.57

(3) 0.37 (4) 0.17

9 Jan Evening

Q4

A particle is executing simple harmonic motion (SHM) of amplitude A, along the x-axis, about $x = 0$. When its potential Energy (PE) equals kinetic energy (KE), the position of the particle will be:

(1) $\frac{A}{2}$ (2) $\frac{A}{2\sqrt{2}}$

(3) $\frac{A}{\sqrt{2}}$ (4) A

9 Jan Evening

Q5

The density of a material in SI unit of is 128 kg m^{-3} In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is:

(1) 40 (2) 16 (3) 640 (4) 410

10 Jan Morning

Q6

A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is:

(1) $\frac{3F}{2mR}$ (2) $\frac{F}{3mR}$

(3) $\frac{F}{2mR}$ (4) $\frac{2F}{3mR}$

10 Jan Morning

Q7

Two stars of masses 3×10^{31} kg each, and at distance 2×10^{11} m rotate in a plane about their common centre of mass O. A meteorite passes through O moving perpendicular to the star's rotation plane. In order to escape from the gravitational field of this double star, the minimum speed that meteorite should have at O is:

(Take Gravitational constant $G = 66 \times 10^{-11}$ Nm^2 kg^{-2})

(1) 2.4×10^4 m/s (2) 1.4×10^5 m/s

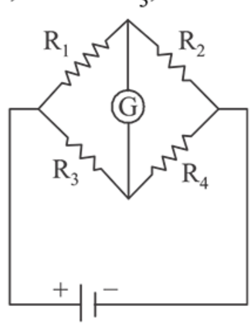
(3) 3.8×10^4 m/s (4) 2.8×10^5 m/s

10 Jan Evening

Q8

The Wheatstone bridge shown in Fig. here, gets balanced when the carbon resistor used as R_1 has the colour code (Orange, Red, Brown). The resistors R_2 and R_4 are 80 Ω and 40 Ω, respectively.

Assuming that the colour code for the carbon resistors gives their accurate values, the colour code for the carbon resistor, used as R_3, would be:



(1) Brown, Blue, Brown (2) Brown, Blue, Black

(3) Red, Green, Brown (4) Grey, Black, Brown

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Q9

An electromagnetic wave of intensity 50 Wm^{-2} enters in a medium of refractive index 'n' without any loss. The ratio of the magnitudes of electric fields, and the ratio of the magnitudes of magnetic fields of the wave before and after entering into the medium are respectively, given by :

(1) $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}\right)$ (2) $\left(\sqrt{n}, \sqrt{n}\right)$

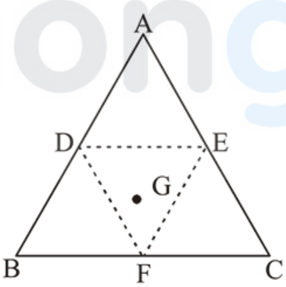
(3) $\left(\sqrt{n}, \frac{1}{\sqrt{n}}\right)$ (4) $\left(\frac{1}{\sqrt{n}}, \sqrt{n}\right)$

11 Jan Morning

Q10

An equilateral triangle ABC is cut from a thin solid sheet of wood. (See figure) D, E and F are the mid-points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0. If the smaller triangle DEF is removed from ABC, the moment of inertia of the remaining figure about the same axis is I.

Then :



(1) $I = \frac{15}{16} I_0$ (2) $I = \frac{3}{4} I_0$

(3) $I = \frac{9}{16} I_0$ (4) $I = \frac{I_0}{4}$

11 Jan Morning

Q11

In an experiment, electrons are accelerated, from rest, by applying a voltage of 500 V. Calculate the radius of the path if a magnetic field 100 mT is then applied.

[Charge of the electron = 1.6×10^{-19} C

Mass of the electron = 9.1×10^{-31} kg]

(1) 7.5×10^{-3} m (2) 7.5×10^{-2} m

(3) 7.5 m (4) 7.5×10^{-4} m

11 Jan Morning

Q12

A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of 20×10^{-6} J/T when a magnetic intensity of 60×10^{3} A/m is applied. Its magnetic susceptibility is:

(1) 3.3×10^{-2} (2) 4.3×10^{-2}
(3) 2.3×10^{-2} (4) 3.3×10^{-4}

11 Jan Evening

Q13

A straight rod of length L extends from $x = a$ to $x = L + a$. The gravitational force it exerts on point mass 'm' at $x = 0$, if the mass per unit length of the rod is $A + Bx^2$, is given by:

(1) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)-BL\right]$

(2) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)-BL\right]$

(3) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)+BL\right]$

(4) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$

12 Jan Morning

Q14

A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original amplitude is close to :

(1) 50s (2) 100s (3) 20s (4) 10s

8 April Evening

Q15

A simple pendulum oscillating in air has period T. The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is :

(1) $2T\sqrt{\frac{1}{10}}$ (2) $2T\sqrt{\frac{1}{14}}$

(3) $4T\sqrt{\frac{1}{15}}$ (4) $4T\sqrt{\frac{1}{14}}$

9 April Morning

Q16

The displacement of a damped harmonic oscillator is given by $x(t) = e^{-0.1t} . \cos(10\pi t + \varphi)$. Here t is in seconds.

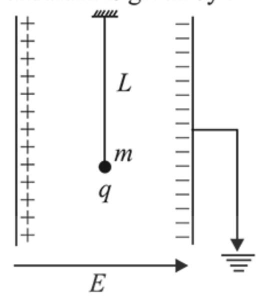
The time taken for its amplitude of vibration to drop to half of its initial value is close to :

(1) 4s (2) 7s (3) 13s (4) 27s

10 April Morning

Q17

A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E, as shown in figure. Its bob has mass m and charge q. The time period of the pendulum is given by :



(1) $2\pi\sqrt{\frac{L}{\left(g+\frac{qE}{m}\right)}}$ (2) $2\pi\sqrt{\frac{L}{\sqrt{g^2-\frac{q^2E^2}{m^2}}}}$

(3) $2\pi\sqrt{\frac{L}{\left(g-\frac{qE}{m}\right)}}$ (4) $2\pi\sqrt{\frac{L}{\sqrt{g^2+\left(\frac{qE}{m}\right)^2}}}$

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SHM - Answers

Q1

(4) Maximum speed is at mean position or equilibrium

At equilibrium position

$$F = kx \Rightarrow x = \frac{F}{k}$$

From work-energy theorem,

$$W_F + W_{sp} = \Delta KE$$

$$F(x) - \frac{1}{2}kx^2 = \frac{1}{2}mv^2 - 0$$

$$F\left(\frac{F}{k}\right) - \frac{1}{2}k\left(\frac{F}{k}\right)^2 = \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2}\frac{F^2}{K} = \frac{1}{2}mv^2$$

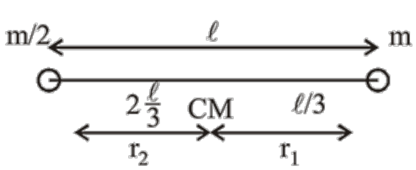
or, $v_{max} = \frac{F}{\sqrt{mk}}$

9 Jan Morning

Q2

(3) As we know, $\omega = \sqrt{\frac{k}{I}}$

$$\omega = \sqrt{\frac{3k}{m\ell^2}} \quad \left[\because \ I_{rod} = \frac{1}{3}m\ell^2\right]$$



Tension when it passes through the mean position,

$$= m\omega^2\theta_0^2\frac{\ell}{3} = m\frac{3k}{m\ell^2}\theta_0^2\frac{\ell}{3} = \frac{k\theta_0^2}{\ell}$$

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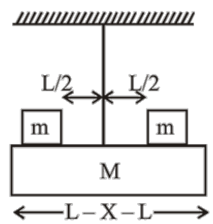
Q3

(3)



$$f_1 = \frac{1}{2\pi}\sqrt{\frac{C}{1}} \quad ...(i)$$

$$= \frac{1}{2}\sqrt{\frac{3C}{ML^2}}$$

$$f_2 = \frac{1}{2\pi}\sqrt{\frac{C}{L^2\left(\frac{M}{3} + \frac{M}{2}\right)}} \quad ...(ii)$$

As frequency reduces by 80%

$$\therefore \ f_2 = 0.8 f_1 \Rightarrow \frac{f_2}{f_1} = 0.8 \quad ...(iii)$$

Solving equations (i), (ii) & (iii)

Ratio $\frac{m}{M} = 0.37$

9 Jan Evening

Q4

(3) Potential energy (U) $= \frac{1}{2}kx^2$

Kinetic energy (K) $= \frac{1}{2}kA^2 - \frac{1}{2}kx^2$

According to the question, U = k

$$\therefore \frac{1}{2}kx^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2$$

$$\Rightarrow x^2 = A^2 \text{ or, } x = \pm\frac{A}{\sqrt{2}}$$

9 Jan Evening

Q5

(1) Density of material in SI unit,

$$=\frac{128\text{kg}}{\text{m}^3}$$

Density of material in new system

$$=\frac{128(50\,\text{g})(20)}{(25\,\text{cm})^3(4)^3}$$

$$=\frac{128}{64}\times(20)=40\,\text{units}$$

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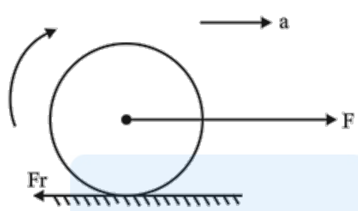
Q6

(4)



$F-\text{fr}=\text{ma}$ ...(i)

$f_r R=I\alpha=\frac{mR^2}{2}\alpha$ ...(ii)

for pure rolling

$a=\alpha R$ ...(iii)

from (i) (ii) and (iii)

$$F-\frac{mR\alpha}{2}=m\alpha R$$

$$F=\frac{3}{2}mR\alpha$$

$$\alpha=\frac{2F}{3mR}$$

10 Jan Morning

Q7

(4) Let M is mass of star m is mass of meteroite
By energy convervation between 0 and ∞.

$$-\frac{GMm}{r}+\frac{-GMm}{r}+\frac{1}{2}mV_{ese}^2=0+0$$

$$\therefore\ v=\sqrt{\frac{4GM}{r}}=\sqrt{\frac{4\times6.67\times10^{-11}\times3\times10^{31}}{10^{11}}}$$

$$\simeq2.8\times10^5\ \text{m/s}$$

10 Jan Evening

Q8

(1) Given, colour code of resistance,
R_1 = Orange, Red and Brown
$\therefore\ R_1=32\times10=320$
using balanced wheatstone bridge principle,

$$\frac{R_1}{R_3}=\frac{R}{R_4}\Rightarrow\frac{320}{R_3}=\frac{80}{40}$$

$\therefore\ R_3=160$ i.e. colour code for R_3 Brown, Blue and Brown

10 Jan Evening

Q9

(3) The speed of electromagnetic wave in free space is given by

$$C=\frac{1}{\sqrt{\mu_0\in_0}} \quad ...(i)$$

In medium, $v=\frac{1}{\sqrt{k\in_0\mu_0}}$ ...(ii)

Dividing equation (i) by (ii), we get

$$\therefore\ \frac{C}{V}=\sqrt{k}=n$$

$$\frac{1}{2}\in_0 E_0^2C=\text{intensity}=\frac{1}{2}\in_0 kE^2v$$

$$\therefore\ E_0^2C=kE^2v$$

$$\Rightarrow\frac{E_0^2}{E^2}=\frac{kV}{C}=\frac{n^2}{n}\Rightarrow\frac{E_0}{E}=\sqrt{n}$$

similarly

$$\frac{B_0^2C}{2\mu_0}=\frac{B^2v}{2\mu_0}\Rightarrow\frac{B_0}{B}=\frac{1}{\sqrt{n}}$$

11 Jan Morning

Q10

(1) Let mass of the larger triangle = M

Side of larger triangle = ℓ

Moment of inertia of larger triangle = ma^2

Mass of smaller triangle = $\frac{M}{4}$

Length of smaller triangle = $\frac{\ell}{2}$

Moment of inertia of removed triangle

$$=\frac{M}{4}\left(\frac{a}{2}\right)^2$$

$$\therefore \frac{I_{removed}}{I_{original}} = \frac{\frac{M}{4}}{M}\cdot\frac{\left(\frac{a}{2}\right)^2}{(a)^2}$$

$$I_{removed} = \frac{I_0}{16}$$

$$\text{So, } I = I_0 - \frac{I_0}{16} = \frac{15I_0}{16}$$

11 Jan Morning

Q11

(4) Radius of the path (r) is given by $r = \frac{mv}{qB}$

$$r = \frac{\sqrt{2mk}}{eB} \quad (\because\ p = mv = \sqrt{2mk})$$

$$= \frac{\sqrt{2meV}}{eB} \quad (\because\ k = eV)$$

$$r = \frac{\sqrt{\frac{2m}{e}V}}{B} = \frac{\sqrt{\frac{2\times9.1\times10^{-31}}{1.6\times10^{-19}}(500)}}{100\times10^{-3}}$$

$$r = \frac{\sqrt{\frac{9.1}{0.16}\times10^{-10}}}{10^{-1}} = \frac{3}{.4}\times10^{-4}$$

$$= 7.5\times10^{-4}$$

11 Jan Morning

Q12

(4) Magnetic susceptibility,

$$\chi = \frac{I}{H}$$

$$\text{where, } I = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{20\times10^{-6}}{10^{-6}}$$

$$= 20\ \text{N/m}^2$$

$$\text{Now, } \chi = \frac{20}{60\times10^3} = \frac{1}{3}\times10^{-3} = 3.3\times10^{-4}$$

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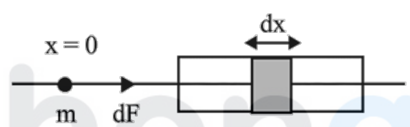
Q13

(4) Given $\lambda = (A + Bx^2)$,
Taking small element dm of length dx at a distance x from x = 0



so, $dm = \lambda\,dx$

$dm = (A + Bx^2)dx$

$$dF = \frac{Gmdm}{x^2}$$

$$\Rightarrow F = \int_a^{a+L}\frac{Gm}{x^2}(A+Bx^2)dx$$

$$= Gm\left[-\frac{A}{x}+Bx\right]_a^{a+L}$$

$$= Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$$

12 Jan Morning

Q14

(3) Time of half the amplitude is = 2*s*
Using, $A = A_0e^{-kt}$

$$\frac{A_0}{2} = A_e\ e^{-k\times2} \qquad \text{...(i)}$$

$$\text{and } \frac{A_0}{1000} = A_e\ e^{-kt} \qquad \text{...(ii)}$$

Dividing (i) by (ii) and solving, we get
$t \simeq 20$ s

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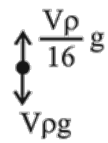
Q15

(3) $T = 2\pi\sqrt{\dfrac{l}{g}}$



$$V\rho\, g_{eff} = V\rho g - \frac{V\rho}{16}g$$

$$g_{eff} = \left(g - \frac{g}{16}\right) = \frac{15g}{16}$$

Now $T' = 2\pi\sqrt{\dfrac{l}{g_{eff}}} = 2\pi\sqrt{\dfrac{l}{\frac{15g}{16}}} = \dfrac{4}{\sqrt{15}}T$

9 April Morning

Q16

(2) Amplitude of vibration at time t = 0 is given by

$$A = A_0 e^{-0.1\times 0} = 1 \times A_0 = A_0$$

also at t = t, if $A = \dfrac{A_0}{2}$

$$\Rightarrow \frac{1}{2} = e^{-0.1t}$$

$t = 10 \ln 2 \simeq 7s$

10 April Morning

Q17

(4) Time period of the pendulum (T) is given by

$$T = 2\pi\sqrt{\frac{L}{g_{\text{eff}}}}$$

$$g_{\text{eff}} = \frac{\sqrt{(mg)^2 + (qE)^2}}{m}$$

$$\Rightarrow g_{\text{eff}} = \sqrt{g^2 + \left(\frac{gE}{m}\right)^2} \Rightarrow T = 2\pi\sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$$

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